



REAL-TIME MONITORING SOLUTION FOR TRANSFORMERS FOR BETTER ASSET MONITORING AND OPTIMIZATION

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Distribution Transformer Monitoring System

Transformer monitoring is conventionally done through periodic inspection and testing of the transformer throughout its lifetime through site visits. This is an inefficient and outdated method. The site person needs to carry various instruments to measure the transformer's key electrical parameters. These parameters include voltage and current, harmonics, oil level and temperature, LV Lug temperature, phase angle, power factor, energy etc. Site personnel are often less skilled, or they do not have the necessary instruments to measure various parameters, and this procedure is time-consuming. The lack of a real-time transformer monitoring system leads to high downtime and transformer burning chances.

The solution to these problems is **real-time monitoring** using direct electrical parameter measurements combined with other leading indicator sensors. The implementation of **real-time monitoring** of the distribution transformer offers benefits such as

- It provides real-time and trend monitoring of transformer condition
- It is a cost-effective means of simultaneously monitoring many transformers



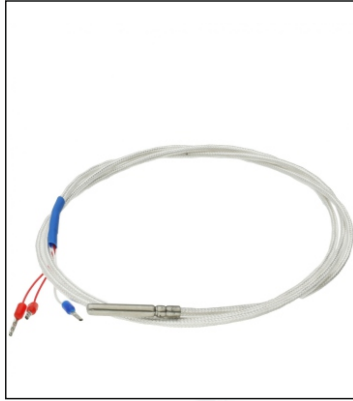
The data monitored is about –

- Three-phase voltage, current, frequency, and phase difference.
- Active, reactive, and apparent power.
- Power factor, THD, voltage, and current harmonics.
- Oil temperature, winding temperature, and oil level.
- Three phase voltage and current
- Frequency and phase angle
- Active power, Reactive power and Apparent power
- Energy consumed in last 24 hours
- Power factor, Voltage THD, Current THD, graphical representation of voltage and current harmonics up to 60
- Phasor Diagram
- LV side lug temperature
- Oil level monitoring in conservator tank
- Main tank oil temperature measurement
- Graphical representation of real-time and historical data
- Predict abnormal conditions of transformers
- Alarm and fault management

The DTMS comes compatible with varied sensor types



**Oil Level
Sensor**



**Oil Temperature
Sensor**



**Current
Sensor**



**Ambient Temperature
Sensor**

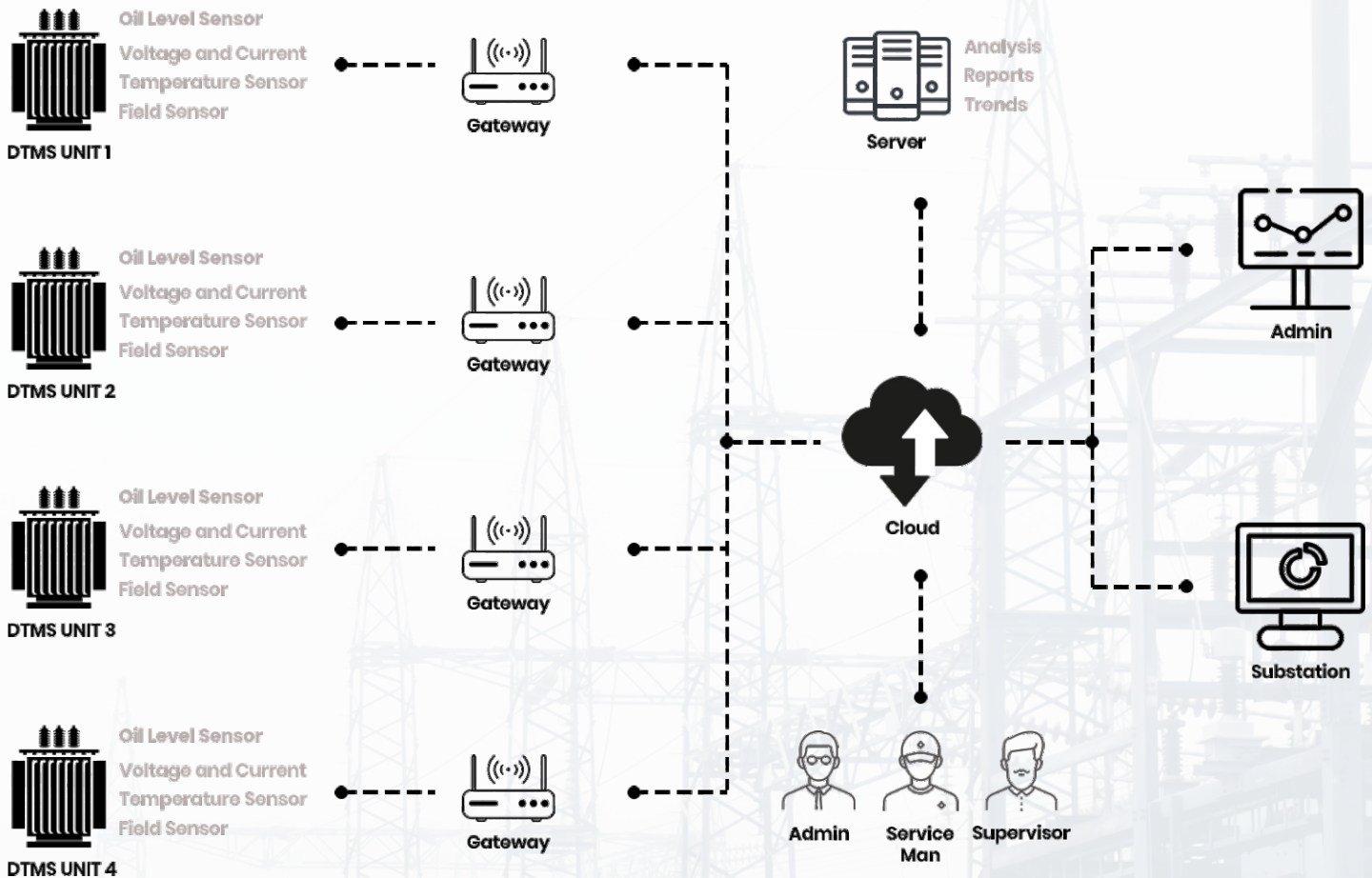
Features

- Supports Rogowski coil sensors (CT) installation without shutdown: reducing downtime. Up to 2500 Amp
- Communication Interfaces: GSM, GPS, Bluetooth, Rs485, and Ethernet.
- Compatible with Three-phase 4-wire system.
- Two inbuilt channels of digital output for potential-free contacts.
- Four channels of optically isolated digital input for field inputs.
- One Li-ion battery with a charging circuit and 4-5 hours of battery backup . Also supports additional 4-20mA type field sensors.

Functions

- Sends all the energy-related data to the central server for remote monitoring and fault identification.
- Integrates various field inputs like humidity sensors, ambient temperature sensors, vibration sensors, oil level measuring sensors, and winding temperature sensors.

DTMS Ecosystem

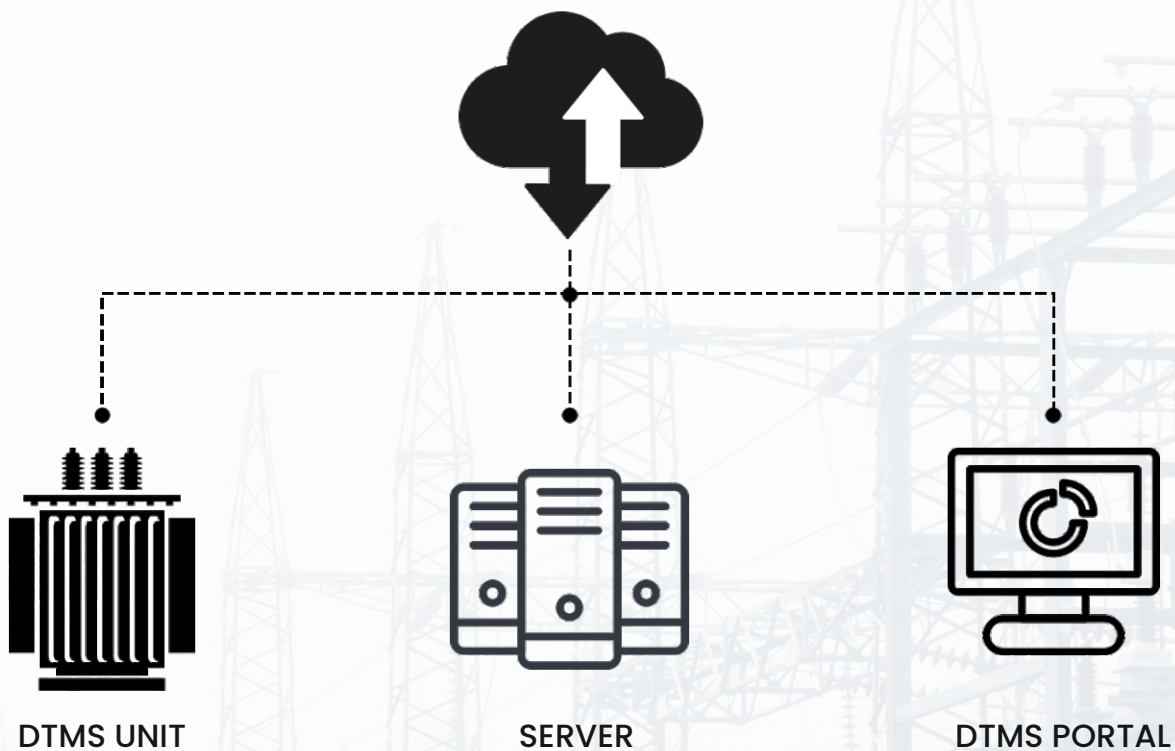


Technical Specifications

- Inbuilt GSM module for communication
- Harmonics measurement. Up to 60 Harmonics
- Analog inputs: 4 channels of 4-20mA
- Digital Inputs: 4 channel - 24 V to 110V & 3 channels for Oil Level Measurement
- Digital Outputs: 2 channels of relay-based outputs up to 10Amp

Smart Firmware

Providing a comprehensive support kit, our DTMS comes equipped with smart firmware and portal assistance.



- DTMS Firmware is developed for measuring energy parameters, monitoring field sensors, and sending all the measured data to the cloud at regular intervals. Data collection, validation, arranging it in a suitable format is one of the most important functions that the firmware serves.
- The firmware handles various arithmetic and logical operations corresponding to the energy data like over-voltage, over-current, and phase imbalance.
- It collects information at unit levels and processes it based on real-time operational requirements.
- It also communicates with the DTMS mobile application, ensuring easy and interactive data transfer at normal operation and maintenance modes.
- It monitors communication happening with the cloud.

Benefits

- Improves transformer reliability.
- Real-time control, as part of a fully integrated, automated power management system.
- Provides the ability to perform calculations to reduce demand charges.
- Reduces DT downtime.
- Measures abrupt increase/decrease in load.
- Compatibility for varied sensor types viz. oil level sensors, oil temperature sensors, palm temperature sensors, ambient temperature sensors, etc.
- No additional power input is required, can take supply from any phase of DT Secondary.
- Reduces technical losses, thus saving power procurement costs.

Smart Portal

- DTMS Portal represents all the monitored and historical data related to energy and other vital data parameters in one place.
- It plays a vital role in the performance analysis of distribution transformers and their corresponding fault management functionalities.
- The portal collects data related to energy and other vital data parameters from the DTMS unit installed at the transformer.

The following functions form the mainstay of the DTMS Smart Portal:

- User Management
- Data Collection & Analysis
- Graphical representation of real-time and historical data
- Alarm Management
- Fault Management
- Transformer Management





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